

Operational Limits of Moore's Law: A Qualitative Synthesis of the Urgency for Alternative Computing Pathways

Input Question

Is there an upper limit to computer processing speed?

Domain

Information Science

Validation Status

needs_data

Problem Statement

The provided context suggests that Moore's Law, an empirical trend of transistor density doubling, is approaching a physical limit due to atomic constraints. The core problem is to determine if an upper limit to processing speed exists based on the 'ultimate limitation' of Moore's Law and whether alternative pathways are required, strictly using the provided evidence which highlights the urgency of new alternatives but does not quantify the limit or validate quantum computing solutions.

Rationale

Moore's Law is identified as an empirical relationship rather than a physical law, yet its 'ultimate limitation' has made the search for alternative pathways urgent (EV-Q091-1f23bf8ec63d855102d37d80). Continuous transistor down-scaling has historically driven information technology development (EV-Q091-1766dc069035172d1e9e4b78), but building smaller transistors with enhanced functionality is now critical to extending these limits (EV-Q091-290a89e4d53a8cc51abb6045). The rationale focuses on synthesizing these qualitative statements to define the current operational limit as the barrier necessitating alternative architectures, rather than a specific numerical speed cap.

Generated Hypotheses

Hypothesis 1

- **Hypothesis**: The current upper limit to computer processing speed is operationally defined by an unspecified 'ultimate limitation' of Moore's Law, which has made the search for alternative pathways an urgent priority over continued traditional scaling.
- **Mechanism**: Historical processing speed improvements were driven by continuous transistor down-scaling (EV-Q091-1766dc069035172d1e9e4b78). However, this trajectory has encountered an 'ultimate limitation' (EV-Q091-1f23bf8ec63d855102d37d80) whose specific nature (physical vs. economic) is not resolved by current evidence. Consequently, the effective upper limit is no longer determined by successful scaling but by

the barrier necessitating alternative pathways. The existence of this limit is inferred from the documented urgency of finding new alternatives rather than from a quantified physical threshold.

- ****Falsifiable Prediction****: If subsequent analysis of allowed evidence demonstrates that traditional transistor down-scaling continues to be the primary driver of performance gains without requiring 'alternative pathways' as an urgent priority, or if the 'ultimate limitation' is explicitly described as non-binding or solved within the provided texts, then this hypothesis is falsified.
- ****Required Observations****: Qualitative confirmation in EV-Q091-1f23bf8ec63d855102d37d80 that the 'ultimate limitation' currently mandates alternative pathways. ; Absence of statements in allowed evidence claiming that traditional scaling remains sufficient for future speed increases. ; Verification that EV-Q091-1766dc069035172d1e9e4b78 frames down-scaling as historical context rather than a continuing solution to the ultimate limitation.
- ****Risk of Being Wrong****: The hypothesis assumes the 'ultimate limitation' is the active constraint defining the current speed ceiling. It is possible the limit exists but is not yet binding, or that the 'urgency' mentioned in EV-Q091-1f23bf8ec63d855102d37d80 is anticipatory rather than reflective of a current hard stop. Additionally, the specific mechanism (physical vs. economic) remains a knowledge_gap.

Hypothesis 2

- ****Hypothesis****: Novel device architectures such as suspended carbon nanotube transistors constitute a validated 'alternative pathway' capable of extending processing speed limits beyond the ultimate limitations of conventional Moore's Law scaling.
- ****Mechanism****: Given that the ultimate limitation of Moore's Law necessitates new pathways (EV-Q091-1f23bf8ec63d855102d37d80), and that building smaller transistors with enhanced functionality is critical to extending these limits (EV-Q091-290a89e4d53a8cc51abb6045), specific alternative implementations like suspended carbon nanotubes are hypothesized to provide the necessary functional extension. This hypothesis posits that CNTs are not just a research topic but a viable candidate for overcoming the stated limitation.
- ****Falsifiable Prediction****: If the content of EV-Q091-290a89e4d53a8cc51abb6045 fails to demonstrate that suspended carbon nanotube transistors offer enhanced functionality relevant to extending Moore's Law limits, or if they are presented solely as theoretical concepts without viability for speed extension, then this specific pathway hypothesis is weakened.
- ****Required Observations****: Explicit linkage in EV-Q091-290a89e4d53a8cc51abb6045 between suspended CNTs and 'extending the limits of Moore's law'. ; Evidence of 'enhanced functionality' in CNTs compared to conventional scaling limits within the allowed text. ; Confirmation that CNTs address the 'ultimate limitation' cited in EV-Q091-1f23bf8ec63d855102d37d80.
- ****Risk of Being Wrong****: EV-Q091-290a89e4d53a8cc51abb6045 may discuss CNTs as one of many options without validating them as the definitive solution. The evidence might only support their potential for 'enhanced functionality' generally, not specifically as the successor to silicon for processing speed. Other alternatives (e.g., Negative Capacitance FETs in EV-Q091-0b57202c1c970ec303188e68) might be equally or more relevant.

Technical Details

This experiment is designed as a qualitative evidence synthesis to validate the hypothesis that the current upper limit to computer processing speed is operationally defined by an unspecified 'ultimate limitation' of Moore's Law, which necessitates alternative pathways. Given the absence of quantitative time-series data in the allowed evidence

cards (EV-Q091-1766dc069035172d1e9e4b78, EV-Q091-1f23bf8ec63d855102d37d80), the methodology shifts from statistical trend fitting to structured textual analysis. The study will extract and categorize statements regarding: (1) the historical role of down-scaling, (2) the existence and nature of the 'ultimate limitation', and (3) the urgency and type of 'alternative pathways'. The mechanism is revised to avoid ungrounded claims about 'fundamental physical constraints', instead treating the limitation as an operational barrier identified in the literature whose specific physical vs. economic nature remains a knowledge gap. The falsifiable prediction is tested by checking if the evidence supports continued scaling without alternatives or if it explicitly mandates new pathways due to this limitation.

Datasets

Source

```
```json
[
{
"name": "Qualitative Evidence Corpus on Moore's Law Limits",
"description": "A structured collection of text segments from allowed evidence IDs describing the status of transistor scaling, the existence of limitations, and the need for alternatives. No quantitative performance metrics are assumed to be present.",
"evidence_ids": [
"EV-Q091-1f23bf8ec63d855102d37d80",
"EV-Q091-1766dc069035172d1e9e4b78",
"EV-Q091-290a89e4d53a8cc51abb6045"
]
}
]
```
```

Target

```
```json
{
"name": "Limitation-Pathway Dependency Matrix",
"description": "A binary/qualitative matrix mapping evidence statements to three categories: 'Scaling Sufficient', 'Ultimate Limitation Exists', and 'Alternative Pathways Urgent'. The target outcome is to verify if the co-occurrence of 'Ultimate Limitation' and 'Alternative Pathways Urgent' is supported while 'Scaling Sufficient' is absent or framed as historical."
}
```

}

...

## Paper Abstract

Background: Moore's Law has historically guided computing power growth through transistor density increases, but it is an empirical trend facing an 'ultimate limitation'. Methods: We conducted a qualitative synthesis of recent arXiv preprints (EV-Q091-1f23bf8ec63d855102d37d80, EV-Q091-1766dc069035172d1e9e4b78, EV-Q091-290a89e4d53a8cc51abb6045) to analyze the narrative surrounding processing speed limits. We coded statements regarding scaling success, limitation existence, and alternative pathway urgency. Validation Plan: The hypothesis that an unspecified 'ultimate limitation' necessitates alternative pathways is tested by verifying the co-occurrence of limitation citations and urgency claims in the evidence, while checking for contradictions suggesting continued scaling sufficiency. Results: pending (待执行验证实验).

## Methods

1. Text Extraction: Extract sentences mentioning 'limit', 'scaling', 'alternative', 'priority', or 'urgent' from allowed evidence IDs. 2. Thematic Classification: Categorize statements into 'Historical Success', 'Ultimate Limitation', or 'Necessity of Alternatives'. 3. Contradiction Check: Identify any claims that traditional scaling remains sufficient without alternatives. 4. Gap Identification: Note if the limitation mechanism (physical/economic) is specified. 5. Hypothesis Validation: Confirm if 'Ultimate Limitation' and 'Urgency of Alternatives' are supported while 'Scaling Sufficient' is restricted to historical context.

## Experiments

### Baselines

```json

[

"Null Hypothesis: Evidence suggests traditional silicon scaling continues to be the primary and sufficient driver for future processing speed increases without urgent need for alternatives.",

"Alternative Hypothesis 1: The 'ultimate limitation' is explicitly defined as a fundamental physical constraint (e.g., quantum tunneling) in the evidence.",

"Alternative Hypothesis 2: The 'ultimate limitation' is explicitly defined as an economic or manufacturing cost barrier in the evidence."

]

...

Metrics

```json

[

"Evidence Support Score for 'Ultimate Limitation' (Count of supporting statements / Total relevant statements)",

"Evidence Support Score for 'Urgency of Alternatives' (Count of supporting statements / Total relevant statements)",

"Mechanism Specificity Index (Binary: 1 if physical/economic mechanism is specified, 0 if unspecified)"

]

...

## Ablation

Remove EV-Q091-290a89e4d53a8cc51abb6045 (CNT specific) to test if the general claim of 'alternative pathways' holds solely based on the general limitation statement in EV-Q091-1f23bf8ec63d855102d37d80 and historical context in EV-Q091-1766dc069035172d1e9e4b78.

## Validation Protocol

Cross-reference all coded statements with their source Evidence IDs. Ensure no external knowledge about Landauer's limit or specific industry roadmaps is introduced. Verify that the 'unspecified' nature of the limitation is accurately reflected if no mechanism is found in the text.

## Results

当前状态：待执行验证实验。系统已生成可复现实验脚本、数据字段清单与评价指标，尚未运行真实实验。

## References

- **\*\*EV-Q091-1f23bf8ec63d855102d37d80\*\*** · arxiv · arXiv:2510.12473
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/2510.12473.pdf> · doi: Not available
- reliability\_note: eligibility\_status=FULLTEXT\_VERIFIED;  
topic\_relevance\_status=DIRECT\_QUESTION\_CORE; locator=page:2|section:page-2|paragraph:1;  
content\_sha256=ff3b574512af6799d3907f3e721fcca0a694d3faf06ae1f9f60ac376982c2412
- **\*\*EV-Q091-290a89e4d53a8cc51abb6045\*\*** · arxiv · arXiv:1607.02612
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/1607.02612.pdf> · doi: Not available
- reliability\_note: eligibility\_status=FULLTEXT\_VERIFIED;  
topic\_relevance\_status=DIRECT\_QUESTION\_CORE; locator=page:1|section:page-1|paragraph:1;  
content\_sha256=65e29bfbd22b12cf564f775219746b95680bc0c997e8f3e2c292a1abff45c438
- **\*\*EV-Q091-1766dc069035172d1e9e4b78\*\*** · arxiv · arXiv:2001.07364
- authors: Not available · year: Not available

- url: <https://arxiv.org/pdf/2001.07364.pdf> · doi: Not available
- reliability\_note: eligibility\_status=FULLTEXT\_VERIFIED;  
topic\_relevance\_status=DIRECT\_QUESTION\_CORE; locator=page:1|section:page-1|paragraph:1;  
content\_sha256=63160c9f9d9a46bbb477d4f3d83a0e89caedf7b452fd48fd24b8ff3eac08de38

## Reviewer Comments

- The revised hypothesis correctly replaces the ungrounded 'fundamental physical constraints' with 'unspecified ultimate limitation' strictly grounded in EV-Q091-1f23bf8ec63d855102d37d80, resolving critical\_issue:fa50eff3b5cc.
- Experiment design has been successfully converted from an invalid quantitative meta-analysis to a qualitative evidence synthesis. The dataset 'Qualitative Evidence Corpus on Moore's Law Limits' is fully supported by allowed evidence IDs, resolving critical\_issue:169503ca97bb.
- Falsifiable prediction now relies exclusively on textual verification within allowed evidence cards rather than external industry roadmaps, satisfying the closed-evidence constraint and resolving critical\_issue:5b2635862b66.
- Knowledge gaps regarding the specific physical vs. economic nature of the limit are explicitly acknowledged in both the hypothesis risk assessment and experiment execution metadata, addressing required\_revision:9a6659c9626f.
- Results field remains 'pending' with no fabrication. Reproducibility checklist is updated to reflect qualitative coding criteria rather than non-existent quantitative metrics.

## Revision History

- auto\_revision\_1: 依据评审意见重做假设与实验设计。

## Reproducibility Checklist

- Verify that only allowed evidence IDs (EV-Q091-1f23bf8ec63d855102d37d80, EV-Q091-1766dc069035172d1e9e4b78, EV-Q091-290a89e4d53a8cc51abb6045) are used for text extraction.
- Confirm that no quantitative gate-length or speed data is assumed to exist in the evidence cards.
- Ensure the hypothesis mechanism does not assert 'fundamental physical constraints' but rather 'unspecified ultimate limitation'.
- Document the classification criteria for 'Historical Context' vs. 'Current/Future Sufficiency' of scaling.
- Explicitly record the 'Mechanism Specificity Index' as 0 if no physical/economic distinction is found in the text.